

$$\omega_1 = \pi c/L, \qquad T = 2L/c,$$

$$u(x,t) = A \cos(\omega_1 t) \sin(\pi x/L),$$

$$E(t) = \tfrac{1}{2} \int_0^L \left[u_t^2 + c^2 u_x^2 \right] dx,$$

$$E'(t) = c^2 [u_t u_x]_0^L = 0,$$

$$E(t) = \frac{A^2 c^2 \pi^2}{4L}.$$